**Sample Project Idea #3**

**Deep Learning for lung cancer diagnosis**

***Brief project description:***

For patients with suspicious nodules in lung, PET/CT is recommended to access for malignancy. Based on PET/CT findings, biopsy is often required to confirm the lung cancer diagnosis. However, biopsy is an invasive procedure and can lead to different side effects such as pneumonia and bleeding. Deep learning has shown promising results for the diagnosis of a variety range of diseases. The goal of this project is to develop deep learning (DL) based models for lung cancer diagnosis from PET/CT, alleviating the need for biopsy such that the risk associated with biopsy can be reduced. Specifically, you will develop DL-based models to classify which cancer type a patient has using PET/CT scans.

***Dataset*:**

The dataset involved in this project can be downloaded from TCIA (https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=70224216 ). This dataset consists of PET/CT scans from 355 patients. The diagnoses based on biopsy include Adenocarcinoma (266), Small Cell Carcinoma (44), Squamous Cell Carcinoma (62) and Large Cell Carcinoma (5).

***Objective:***

The objective is to develop deep learning methods to diagnose the first three types of cancers. The detailed description of this dataset can be found on the same webpage provided earlier.